

Demonstration of Proposed Features

Date	04 july 2026
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document presents the key features proposed during the planning phase of the Human Development Index (HDI) Prediction Using Machine Learning project and their implementation status. It demonstrates that all planned functionalities were successfully developed, tested, and demonstrated.

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	HDI Prediction	Predicts the Human Development Index using four	Implemented	Yes	Working successfully
2	User-Friendly Web Interface	HTML/CSS interface for entering input values and	Implemented	Yes	Easy to use
3	Linear Regression Model	Trained machine learning model for HDI	Implemented	Yes	Model saved as model.pkl
4	HDI Category Classification	Displays HDI category (Very High, High, Medium, Low).	Implemented	Yes	Accurate categorization
5	Data Visualization	Scatter plot, heatmap, and HDI distribution graph for	Implemented	Yes	Used during EDA
6	Flask Deployment	Integrated model with Flask for real-time prediction.	Implemented	Yes	Runs successfully on local server

Feature Implementation Summary

Item	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation	100%

Conclusion

All proposed features of the Human Development Index (HDI) Prediction Using Machine Learning project were successfully implemented and demonstrated. The application accurately predicts the HDI score, classifies the result into the appropriate HDI category, provides a user-friendly interface, and integrates the trained machine learning model with a Flask web application. The project achieved a 100% implementation rate, satisfying all planned functional requirements.